Ultra-low power consumption high-performanc e 2.4GHz GFSK wireless transceiver chip

Main Features

Works in the 2.4GHz ISM frequency band.

Modulation mode: GFSK/FSK

Data rate: 2Mbps/1Mbps/250Kbps

☒ Ultra-low shutdown current: 1uA

☒ Ultra-low standby current: 15uA

Receiving sensitivity: -83 dBm @ 2 Mbps

☒ Maximum transmission power: 7 dBm

Receiving current (2Mbps): 15mA

Transmit current (2Mbps): 12mA (0dBm)

☒ Integrated high PSRR LDO internally

☒ Wide power supply voltage range: 1.9 - 3.6V

☒ Analog and digital I/O voltage range: 1.9 - 5.25V

☒ Quick start-up time: ⩽ 130 µs

☒ Up to 10 MHz four-wire SPI interface

☒ Integrated intelligent ARQ baseband protocol engine internally

Hardware interrupt output for data transmission and reception

☒ Supports 1-bit RSSI output

☒ Low-cost crystal oscillator: 16MHz ± 60ppm

☒ Minimal peripheral components, reducing system application costs

☒ QFN20 package or COB package

Application scope

Wireless mouse and keyboard

◆Wireless remote control, motion-sensing devices

Active RFID

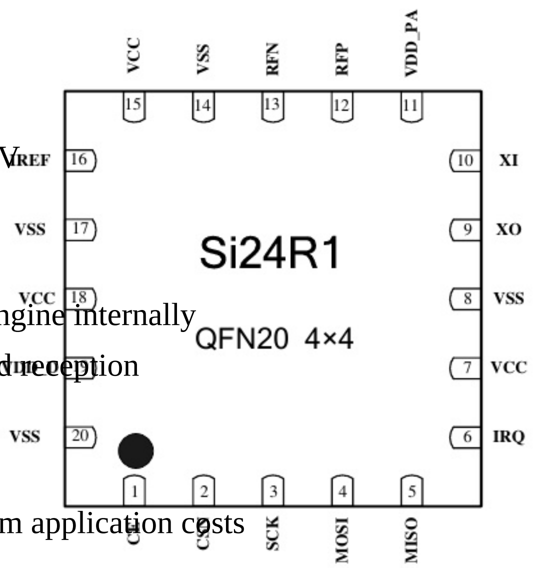
Smart grid, smart home

Wireless audio

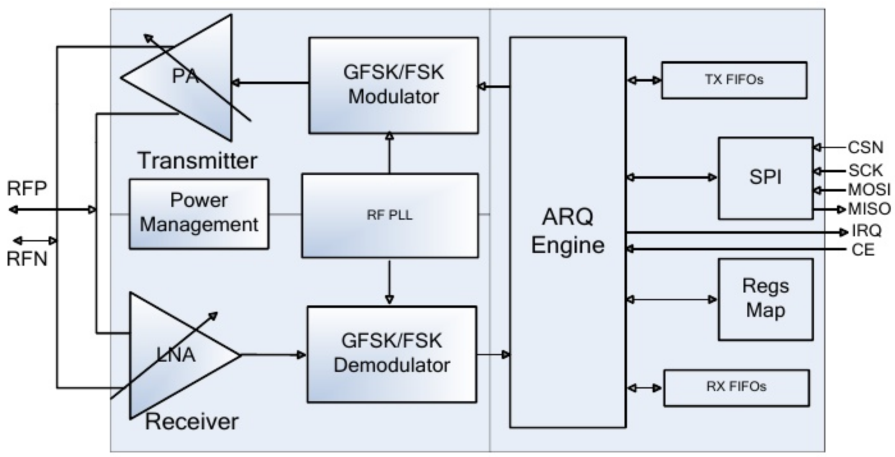
Wireless data transmission module

Low-power self-organizing wireless sensor network node

Encapsulation diagram



Structural block diagram



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Abbreviations of Terms

Terminology	Description	Please prov ide the Chin ese text for translation.
ARQ	Auto Repeat-reQuest	Automatic Retransmission Request
ART	Auto ReTransmission	Automatic retransmission
ARD	Auto Retransmission Delay	Automatic Repeat Request Delay
BER	Bit Error Rate	Bit error rate
CE	Chip Enable	Chip enable
CRC	Cyclic Redundancy Check	Cyclic Redundancy Check
CSN	Chip Select	Chip selection
DPL	Dynamic Payload Length	Dynamic carrier length
GFSK	Gaussian Frequency Shift Keying	Gaussian frequency shift keying
IRQ	Interrupt Request	Interrupt request
ISM	Industrial-Scientific-Medical	Industry - Science - Medicine
LSB	Least Significant Bit	Least significant bit
Mbps	Megabit per second	megabits per second
MCU	Micro Controller Unit	microcontroller
MHz	Mega Hertz	megahertz
MISO	Master In Slave Out	Host input, slave output
MOSI	Master Out Slave In	Host output, slave input
MSB	Most Significant Bit	Most significant bit
PA	Power Amplifier	Power amplifier
PID	Packet Identity	Data packet identification bit
PLD	Payload	carrier wave
RX	RX	receiving end
TX	TX	transmitter end
PWR_DWN	Power Down	Power failure
PWR_UP	Power UP	Power on
RF_CH	Radio Frequency Channel	Radio frequency channel
RSSI	Received Signal Strength Indicator	Signal strength indicator
RX	Receiver	receiver
RX_DR	Receive Data Ready	Data reception is ready.
SCK	SPI Clock	SPI clock
SPI	Serial Peripheral Interface	Serial Peripheral Interface
TX	Transmitter	transmitter
TX_DS	Transmit Data Sent	Sent data
XTAL	Crystal	crystal oscillator

Contents

1 Introduction

The Si24R1 is a 2.4GHz ISM band wireless transceiver chip designed specifically for low-power wireless applications, featuring an integrated embedded ARQ baseband protocol engine. It operates within the frequency range of 2400MHz to 2525MHz and offers 126 channels each with a 1MHz bandwidth. The chip incorporates a high PSRR LDO power supply, ensuring stable operation within a wide power supply range of 1.9V to 3.6V.

The Si24R1 adopts GFSK/FSK digital modulation and demodulation technology. The data transmission rate can be adjusted, supporting three data rates: 2Mbps, 1Mbps, and 250Kbps. A higher data rate can complete the same data transmission and reception in a shorter time, thus achieving lower power consumption. The output power of the chip is adjustable, and the appropriate output power can be configured according to the actual application scenario to save system power consumption.

The Si24R1 has been specially optimized for low-power applications. In the off mode, all register values and FIFO values remain unchanged, with an off current of 1uA. In the standby mode, the clock remains operational, with a current of 15uA, and data transmission and reception can commence within a maximum of 130uS.

The Si24R1 is easy to operate. Only a few registers of the chip need to be configured through the SPI interface by the MCU to achieve data transmission and reception. The embedded ARQ baseband engine is based on the packet communication principle and supports multiple communication modes. It can operate the ARQ protocol manually or automatically. The internal integrated transmit and receive FIFO ensures continuous data transmission between the chip and the MCU. The enhanced ARQ baseband protocol engine can handle all high-speed operations, significantly reducing the system consumption of the MCU.

Si24R1 features an extremely low system application cost. It only requires one MCU and a few peripheral passive components to form a wireless data transmission and reception system. The digital I/O is compatible with multiple standard I/O voltages such as 2.5V, 3.3V, and 5V, allowing direct connection to various MCU ports.

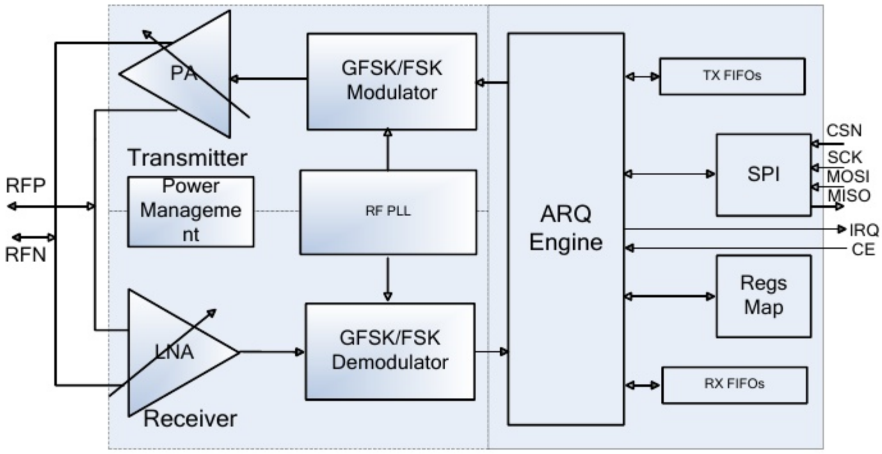

Figure 1-1 Block Diagram of Chip Structure

2 Pin Information

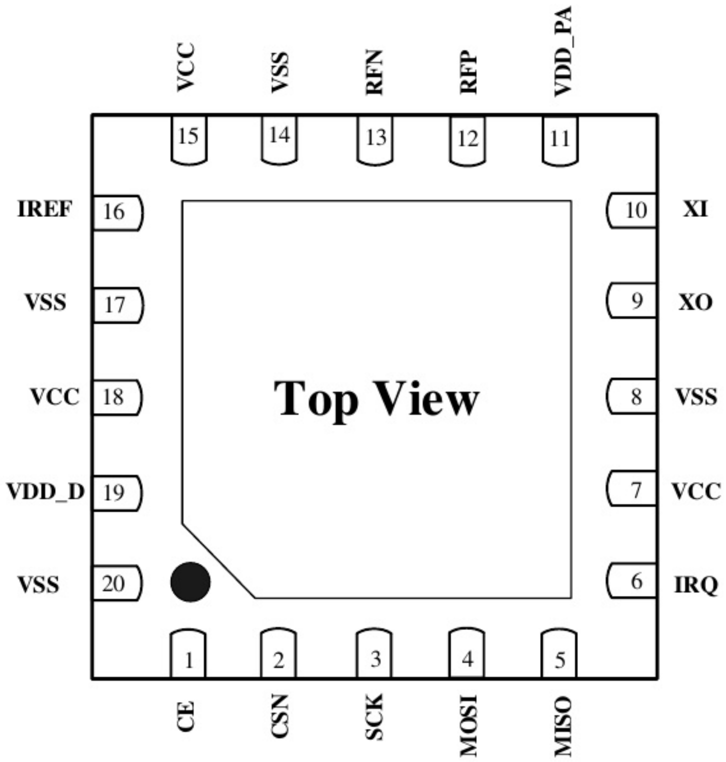

Figure 2-1 Top perspective view of Si24R1 (QFN20 4×4)

Table 2.1 Pin Function Description

port	Port name	Port type	Function Description
1	CE	DI	Chip enable signal, activates RX or TX mode
2	CSN	DI	SPI chip select signal
3	SCK	DI	SPI clock signal
4	MOSI	DI	SPI input signal
5	MISO	DO	SPI output signal
6	IRQ	DO	Maskable interrupt signal, active low
7, 15, 18	VCC	Power	Power supply (+1.9 ~ +3.6V, DC)
8, 14, 17, 20	VSS	Power	Ground (0V)
9	XO	AO	Output pin of crystal oscillator
10	XI	AI	Crystal oscillator input pin
11	VDD_PA	Power	The power output pin (+1.8V) for powering the built-in PA
12	RFP	RF	Antenna interface 1
13	RFN	RF	Antenna interface 2
16	IREF	AI	Reference current

19	VDD_D	PO	The internal digital circuit power supply must be connected to a decoupling capacitor.
	Die exposed	Power	Ground (0V), it is recommended to be connected to the large area ground of the PCB.

3 Working mode

3.1 State transition diagram

The Si24R1 chip has an internal state machine that controls the transitions of the chip between different working modes.

The Si24R1 can be configured in five operating modes: Shutdown, Standby, Idle-TX, TX, and RX. The state transition diagram is shown in Figure 3-1.

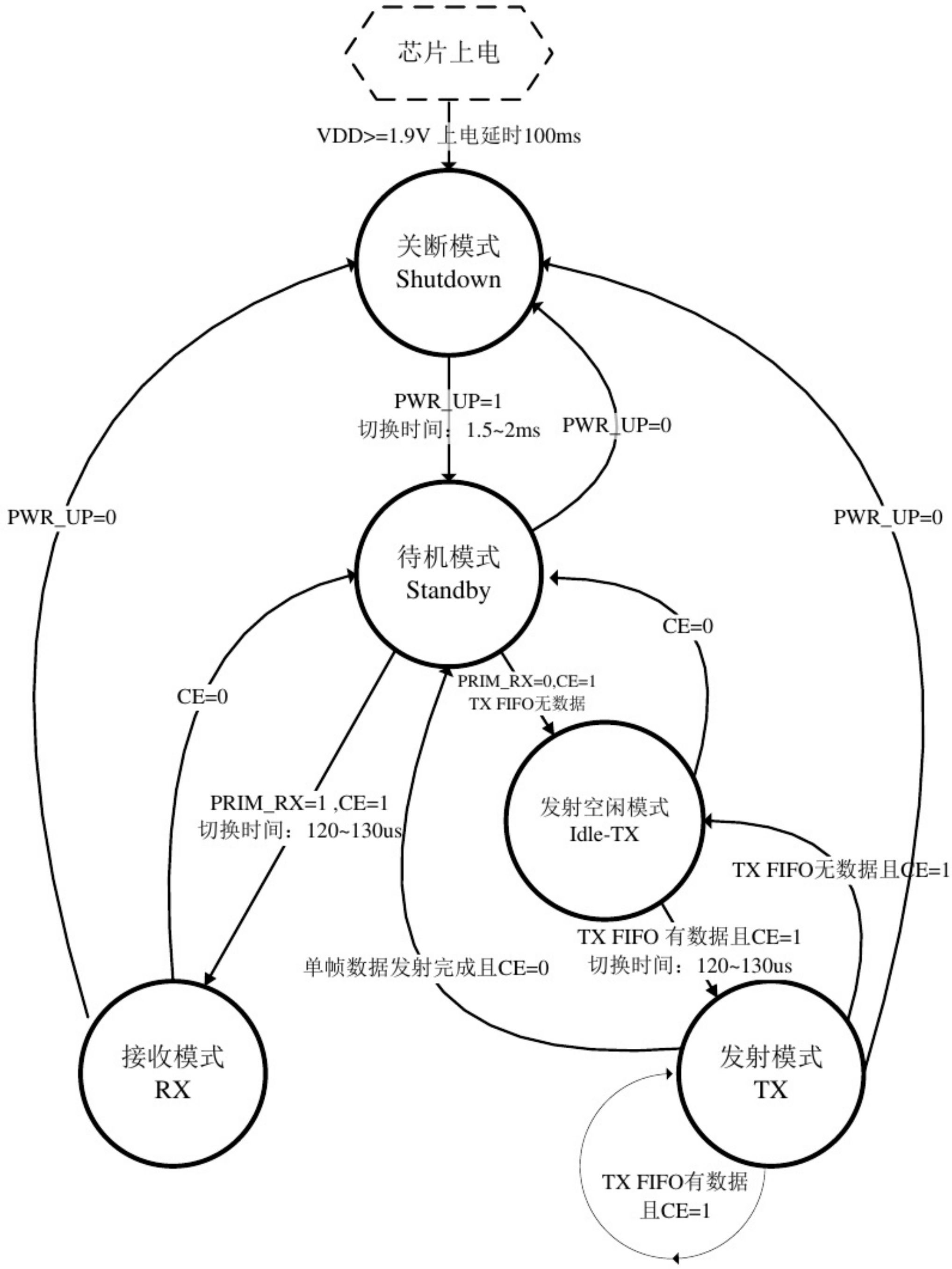

Figure 3-1 Si24R1 Working Mode Switching Diagram

3.1.1 Shutdown working mode

In the Shutdown working mode, all the transceiver function modules of Si24R1 are turned off, and the chip stops working, consuming the minimum current. However, all internal register values and FIFO values remain unchanged and can still be read and written through SPI. Setting the value of the PWR_UP bit in the CONFIG register to 0 immediately returns the chip to the Shutdown working mode.

3.1.2 Standby 工作模式 -> 3.1.2 Standby Working Mode

In Standby working mode, only the crystal oscillator circuit is active, ensuring that the chip can start up quickly while consuming less current. Set the value of the PWR_UP bit in the CONFIG register to 1, and the chip will enter Standby mode after the clock stabilizes. The clock stabilization time of the chip is generally 1.5 to 2 ms, which is related to the performance of the crystal oscillator. When the CE pin is 1, the chip will transition from Standby mode to Idle-TX or RX mode. When CE is 0, the chip will return from Idle-TX, TX or RX mode to Standby mode.

3.1.3 Idle-TX Operating Mode

In Idle-TX working mode, the crystal oscillator circuit and the clock circuit are operational. Compared to Standby mode, the chip consumes more current. When the TX FIFO register at the transmitting end is empty and the pin CE = 1, the chip enters Idle-TX mode. In this mode, if a new data packet is sent to the TX FIFO, the internal circuits of the chip will immediately start and switch to TX mode to send the data packet.

In Standby and Idle-TX working modes, all internal register values and FIFO values remain unchanged and can still be read from and written to via SPI.

3.1.4 TX Working Mode

When data needs to be sent, the device must switch to TX working mode. The conditions for the chip to enter TX working mode are as follows: there is data in the TX FIFO, the value of the PWR_UP bit in the CONFIG register is 1, the value of the PRIM_RX bit is 0, and there is a high pulse of at least 10us on the CE pin. The time for the Idle-TX mode to switch to TX mode is between 120us and 130us, but will not exceed 130us. After a single packet of data is sent, if CE=1, the working mode of the chip is determined by the status of the TX FIFO. If there is still data in the TX FIFO, the chip remains in TX working mode and sends the next packet of data; if there is no data in the TX FIFO, the chip returns to Idle-TX mode. If CE=0, it immediately returns to Standby mode. After the data transmission is completed, the chip generates a data transmission completion interrupt.

3.1.5 RX Working Mode

When data needs to be received, the chip should be switched to the RX working mode. The conditions for the chip to enter the RX working mode are: setting the value of the PWR_UP bit in the CONFIG register to 1, the value of the PRIM_RX bit to 1, and the CE pin to 1. The time for the chip to switch from Standby mode to RX mode is 120 to 130 microseconds. When the address of the received data packet matches the chip's address and the CRC check is correct, the data will be automatically stored in the RX FIFO and a data reception interrupt will be generated. The chip can store up to three valid data packets simultaneously. When the FIFO is full, the received data packets will be automatically discarded.

In receive mode, the power of the received signal can be detected through the RSSI register. When the received signal strength is greater than -60 dBm, the value of the RSSI bit in the RSSI register will be set to 1. Otherwise, RSSI = 0. There are two methods to update the RSSI register: RSSI will be automatically updated after receiving a valid data packet, and it will also be automatically updated when the chip is switched from RX mode to Standby mode. The value of RSSI will change with temperature.

The range is within ±5 dBm.

4 Data Packet Processing Protocol

Si24R1 is based on packet communication and supports the stop-and-wait ARQ protocol. The ARQ protocol baseband processing engine inside the chip can automatically handle ACK and NO_ACK packets without the intervention of an external microcontroller. The ARQ protocol baseband processing unit supports dynamic data lengths from 1 to 32 bytes, with the data length specified within the data packet. It can also use a fixed data length, which is set through registers. The baseband processing unit automatically unpacks, packs, replies with ACK confirmation signals, and retransmits data. It has six communication channels internally, directly supporting a 1:6 star network.

4.1 ARQ Packet Format

A complete ARQ data packet consists of a preamble, address, packet control word, payload data, and CRC. As shown in Figure 4-1, this constitutes a complete packet.

前导码	地址	包控制字	负载数据	CRC

Figure 4-1 A complete ARQ packet with data

The preamble field is mainly used for data synchronization upon reception. It is automatically appended by the chip during transmission and removed by the chip upon reception, and is not visible to the user.

The address field is the address of the data receiving party. Only when this address matches the address in the address register of the chip will the data be received. The address length can be configured as 3, 4, or 5 bytes through the configuration register AW.

It should be noted that the highest byte of the address cannot be set to 0xFF, 0x00, 0xA5, 0x5A, 0xAA, or 0x55; otherwise, it may lead to reception failure.

The length of the packet control field is 9 bits, and its structure is shown in Figure 4-2.

数据包长度 6bit	PID 2bit	NO_ACK 1bit

Figure 4-2 Format of the Packet Control Field

The subfield of data packet length specifies the length of the data packet, which can range from 0 to 32 bytes.

For example: 000000 = 0 byte (The packet is empty)

100000 = 32 bytes (The data packet length is 32 bytes.)

The PID subfield informs the receiving end whether the packet is a new one or a retransmitted one, which can prevent the receiving end from receiving the same packet multiple times. The transmitting end writes to the FIFO via SPI, and the value of PID is automatically incremented. The receiving end compares the PID and CRC to determine if the received packet is a new one or a retransmitted one. If the PID is the same as that of the previous packet, the CRC is compared. If the CRC is also the same, it is judged as a retransmission of the previous frame of data and the data is discarded.

When the NO_ACK subfield is 1, it indicates that the transmitter informs the receiver that an ACK confirmation signal is not required. For the transmitter, setting the NO_ACK bit to 1 requires first configuring the EN_DYN_ACK bit in the FEATURE register to 1.

Use the W_TX_PAYLOAD_NOACK command to write to the FIFO. When a packet of this kind is received, the receiving end will not send an ACK confirmation signal to the transmitting end. (Even if the receiving end is in the ACK receiving mode.)

The load data field is the content of the transmitted data and can be up to 32 bytes in length.

The CRC field is the CRC value of the packet. CRC supports two types: 8-bit and 16-bit. The length of CRC is configured through the CRCO bit in the CONFIG register.

4.2 ARQ Communication Mode

In TX mode, the sender automatically packages the preamble, address, packet control word, payload data, and CRC. The signal is then modulated through the RF module and transmitted via the antenna.

In RX mode, the receiving end continuously detects valid addresses in the demodulated signal. Once an address matching the receiving address is detected, data reception begins. If the received data is valid, the payload data is stored in the RX FIFO and an interrupt is generated to notify the MCU. The MCU can access the RX FIFO register at any time through the SPI interface to read the data.

4.2.1 ACK mode

When writing data to the TX FIFO of the transmitter using the W_TX_PAYLOAD command, the NO_ACK flag bit in the packet control field of the data packet is reset after the data is packed. After the receiver receives a valid frame of data and generates an RX_DR interrupt, it will automatically send an ACK signal. When the transmitter receives the ACK signal, it will automatically clear the data in the TX FIFO and generate a TX_DS transmission interrupt, indicating that the communication was successful.

When the receiving end sends the ACK signal, it uses the address of the receiving pipe as the destination address. Therefore, the sending end needs to set the address of receiving pipe 0 to be the same as its own sending address so as to receive the ACK signal.

If the sender does not receive the ACK signal within the ARD time, it will resend the previous frame of data. When the number of retransmissions reaches the maximum and no confirmation signal is received, the sender generates a MAX_RT interrupt. The MAX_RT interrupt cannot be cleared until the next data transmission is initiated. All interrupts are cleared by writing to the status register. The PLOS_CNT register increments by 1 each time a MAX_RT interrupt occurs, used to record the number of lost data packets in the current frequency band. The ARC_CNT register records the current number of data retransmissions and is reset when a new data packet is sent. The maximum number of retransmissions and the ARD time are configured through the SETUP_RETR register. The receiver's automatic ACK signal reply is enabled by the EN_AA register.

Figure 4-3 shows the completion of a communication in ACK mode.

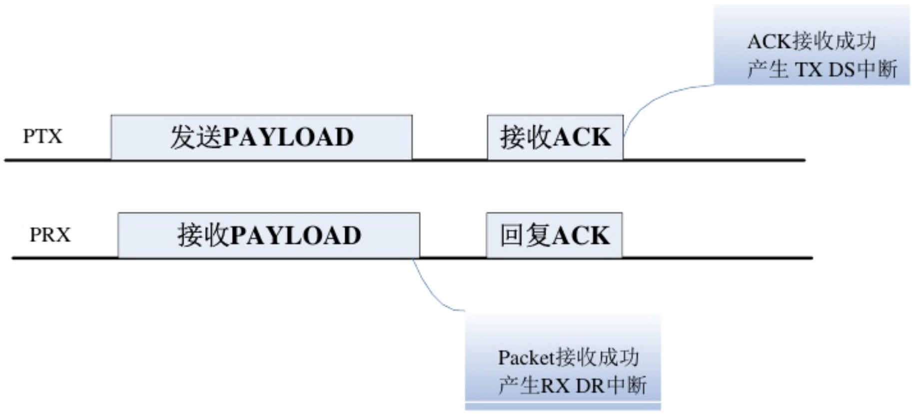

Figure 4-3 ACK Communication Mode

Each time the sender transmits a new data packet, the corresponding PID is automatically incremented by 1. Therefore, the PIDs of two consecutive data packets sent should be different. If several consecutive frames of data are lost in the link, the PIDs of two consecutive frames of data received by the receiver may be the same.

If the receiving end finds that the PID of the received data is the same as that of the previous frame, it will compare the CRC. If the CRC is also the same, it will be judged as a retransmission of the previous frame data, and the data will be discarded. Then, an ACK signal will be re-sent. In Figure 4-4, the first data transmission from the sending end did not receive an ACK signal. After retransmission, an ACK signal was received, and the data communication was successfully completed.

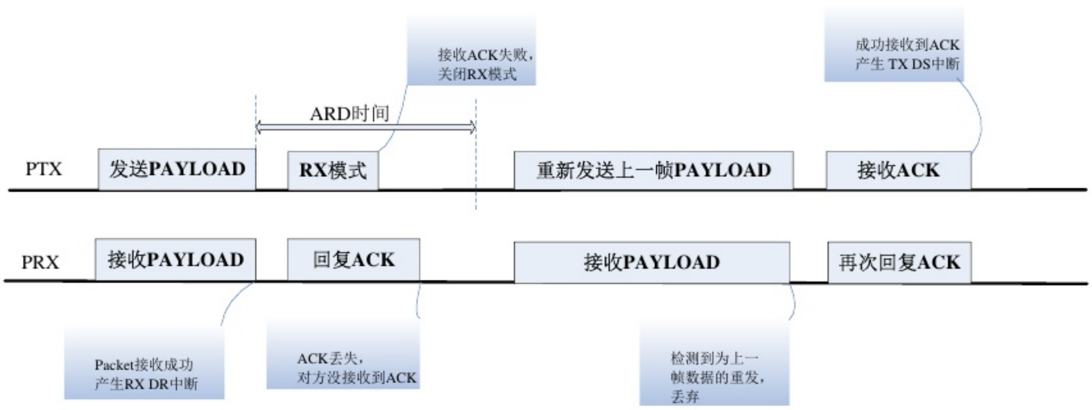

Figure 4-4 Communication Mode without ACK PAYLOAD

When the receiving end replies with an ACK signal, it can simultaneously send an ACK signal with payload data (ACKPAYLOAD). To enable this function, the EN_ACK_PAY bit in the FETURE register needs to be configured, and both sides must enable dynamic payload length.

The receiving end first writes the corresponding ACKPAYLOAD to the TX FIFO using W_ACK_PAYLOAD for the receiving data pipeline. When a new valid frame is received in this pipeline, the RX_DR interrupt is generated, and an ACK is automatically sent back, with the ACKPAYLOAD automatically packaged and sent to the transmitting end. When the transmitting end receives the ACK signal with the payload data, both the TX_DS and RX_DR interrupts are generated. When the receiving end receives another valid data packet from the transmitting end again, it indicates that the transmitting end has received the ACKPAYLOAD, and the data in the TX FIFO is cleared, generating both the RX_DR and TX_DS interrupts. If the received data is a retransmission of the previous packet, this process is repeated.

ACKPAYLOAD is packaged and sent out as an ACK signal. In Figure 4-5, after the first transmission by the sender, no ACK signal with ACKPAYLOAD is received, so it is resent. The receiver repackages this ACKPAYLOAD and, upon receiving it, proceeds to send the next frame of data.

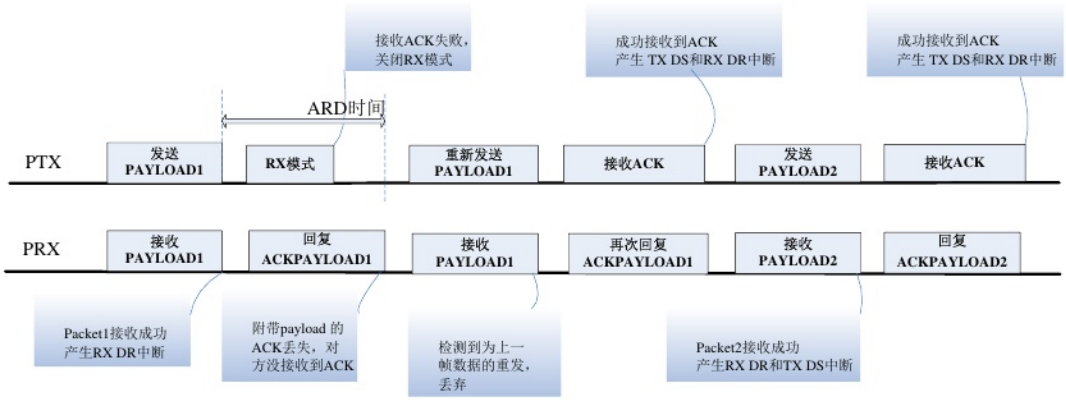

Figure 4-5 Communication Mode with ACK PAYLOAD

4.2.2 NO ACK mode

When the W_TX_PAYLOAD_NOACK command is used to write the TX PAYLOAD on the sender side, the NO_ACK bit in the data packet is set to 1. After the sender finishes sending a packet of data, it immediately generates a TX_DS interrupt and starts preparing to send the next packet. Upon receiving the data, the receiver checks that the NO_ACK bit is set and the data is valid, then generates an RX_DR interrupt. At this point, the data communication for one frame is completed, and no ACK signal is sent back. The W_TX_PAYLOAD_NOACK command is enabled through the EN_DYN_ACK bit in the FEATURE register.

4.2.3 Dynamic PAYLOAD Length and Static PAYLOAD Length

The sender enters the dynamic payload length mode by configuring the EN_DPL bit in the FEATURE register and the DPL_P0 bit in the DYNPD register. The first 6 bits of the packet control field in the sent data packet represent the length of the data to be sent.

When the EN_DPL bit in the FEATURE register at the receiving end is configured and the dynamic enable of the corresponding pipe in the DYNPD register is enabled, data is automatically received based on the data length in the packet control word of the data packet. Therefore, the length of the payload data received each time can be different, and the length of the payload data can be read out through the R_RX_PL_WID command. If the default is a static payload length, the length of the payload data transmitted by the sender each time must be the same and consistent with the value of the RX_PW_Px register configured in advance by the receiver.

4.2.4 Multi-channel communication

The transceiver can simultaneously perform bidirectional or unidirectional communication between 6 transmitting ends and 1 receiving end. At this time, the receiving end needs to enable each pipe in the EN_RXADDR register and set the address of each receiving pipe to be the same as the sending address of the corresponding transmitting end. Among them, receiving pipe 0 has a separate 5-byte address, while pipes 1-5 share the upper 4 bytes effectively.

Address.

If the transmitting end needs to receive the ACK signal, it also needs to set the address of its receiving pipe 0 to be the same as its own sending address.

In the multi-pipe communication mode, the address settings for the sender and receiver are as shown in Figure 4-6.

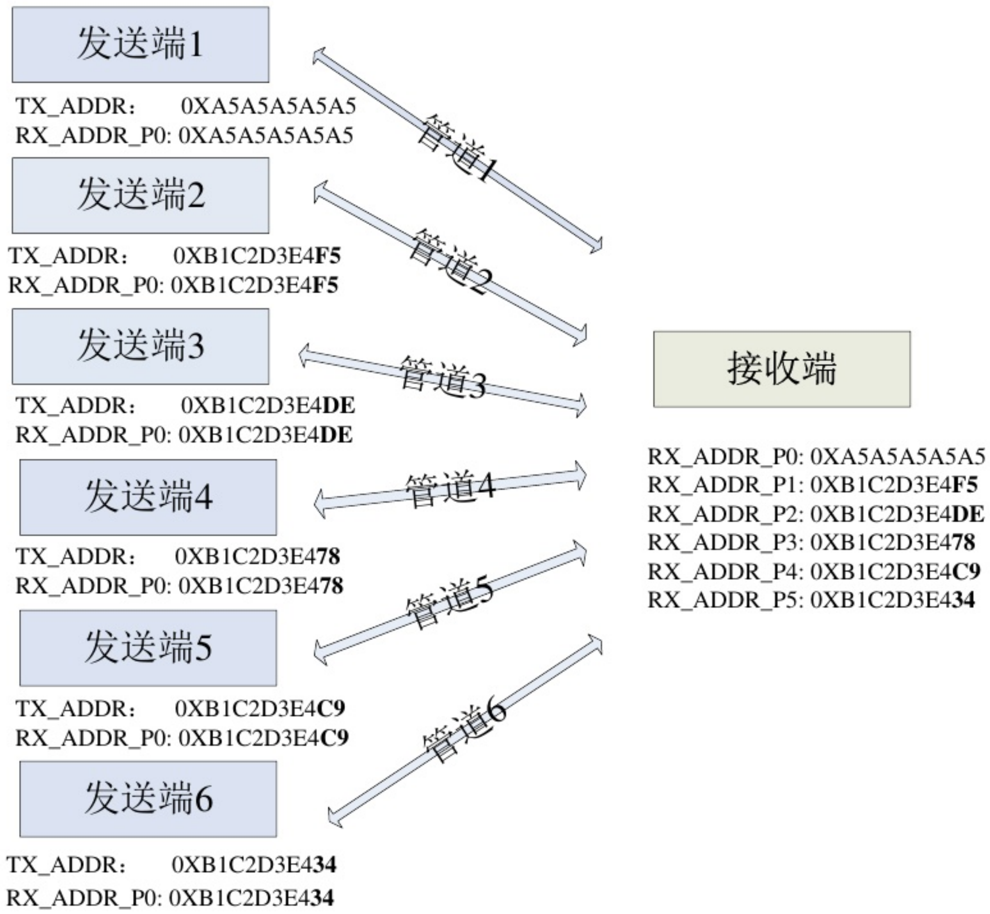

Figure 4-6 Multi-pipe Communication Mode

Multi-channel operation can directly support a star network with a maximum ratio of 1:6.

5 SPI Data and Control Interface

The chip adopts a standard four-wire SPI interface, with a measured maximum read/write speed exceeding 10 Mb/s. The external microcontroller can configure the chip through the SPI interface, including reading and writing function registers, reading and writing FIFO, reading the chip status, and clearing interrupts, etc.

5.1 SPI Command

The SPI commands are listed in Table 5-1. When CSN transitions from high to low, the SPI interface starts to work. For each SPI operation, the first byte output on MISO is the value of the status register. Whether to output values or not (high impedance state if not output) is determined by the command. In the command format, the command word is input from the MSBit to the LSBit. In the data format, it is input from the LSByte to the MSByte, and within each byte, from the MSBit to the LSBit. For details, please refer to the SPI timing diagrams, Figures 5-1 and 5-2.

Table 5-1

Command name	Command word (binary)	# Data bytes	Operation
R_REGISTER	000A AAAA	1 to 5 LSByte first	Read register command, AAAAA represents the register address (refer to the register table).
W_REGISTER	001A AAAA	1 to 5 LSByte first	Write register command, AAAAA represents the register address (refer to the register table). Operations are only allowed in Shutdown, Standby, and Idle-TX modes.
R_RX_PAYLOAD	0110 0001	1 to 32 LSByte first	Read the received data from the FIFO, 1 to 32 bytes. After reading, the data in the FIFO is deleted. This is applicable to the receiving mode.
W_TX_PAYLOAD	1010 000	1 to 32 LSByte first	Write the transmission load data, with a size of 1 to 32 bytes, applicable to the transmission mode.
FLUSH_TX	1110 0001	0	Clear the TX FIFO, applicable to the transmission mode.
FLUSH_RX	1110 0010	0	Clear the RX FIFO, which is applicable in the receiving mode. If an ACK needs to be sent, the FIFO cannot be cleared before the ACK operation is completed; otherwise, it will be regarded as a communication failure.
REUSE_TX_PL	1110 001	0	This command cannot be used by the sender after emptying the TX FIFO or writing new data to the FIFO.
R_RX_PL_WID	0110 000	1	Read the number of received data bytes.
W_ACK_PAYLOAD	1010 1PPP	1 to 32 LSByte first	For the recipient, data is sent out in the form of ACK via the PIPE PPP, and a maximum of three frames of data are allowed to be stored in the FIFO.
W_TX_PAYLOAD_NO ACK	1011 000	1 to 32 LSByte first	This command is applicable to the transmission mode. When using this command, the AUTOACK position should also be set to 1.
NOP	1111 111	0	No operation. It can be used to return the STATUS value.

5.2 SPI Timing

SPI operations include basic read and write operations as well as other command operations. The timing diagrams are shown in Figures 5-1 and 5-2.

注：只能在 Shutdown、Standby 和 Idle-TX 模式下才能对寄存器进行配置。

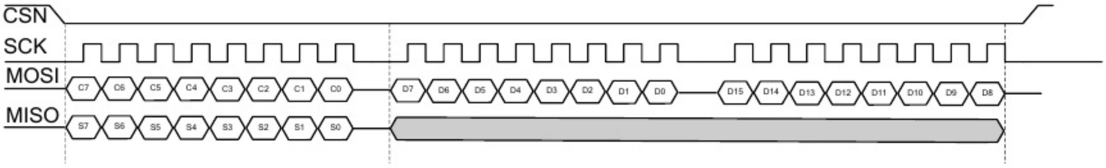

图 5-1 SPI 写操作

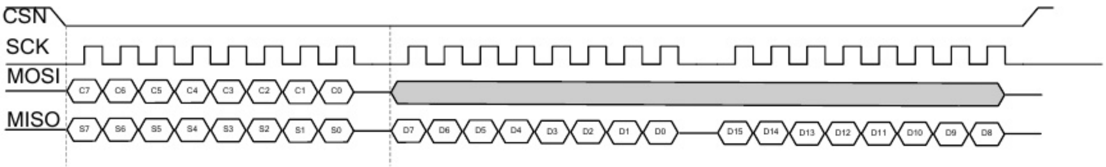

图 5-2 SPI 读操作

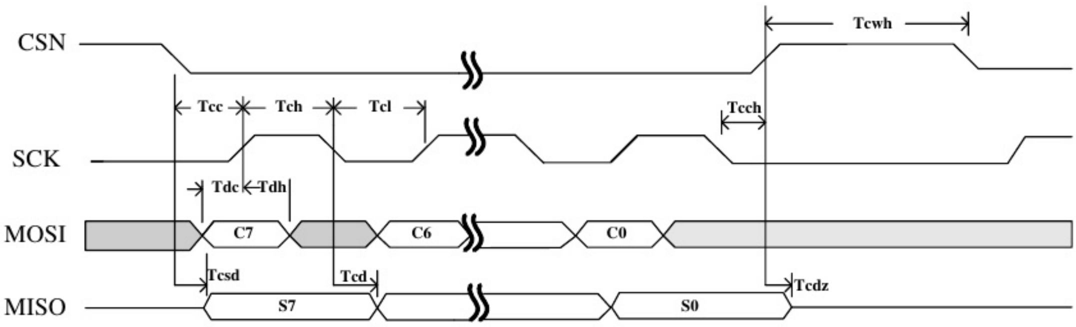

图 5-3 SPI 典型时序

Table 5-1 shows the typical timing parameters of SPI.

Table 5-1 SPI Timing Parameters

Symbol	Parameters	Min	Max	Units
Tdc	Data to SCK Setup	2		ns
Tdh	SCK to Data Hold	2		ns
Tcsd	CSN to Data Valid		42	ns
Tcd	SCK to Data Valid		58	ns
Tcl	SCK Low Time	40		ns
Tch	SCK High Time	40		ns
Fsck	SCK Frequency	0	10	MHz
Tr,Tf	SCK Rise and Fall		100	ns
Tcc	CSN to SCK Setup	2		ns
Tcch	SCK to CSN Hold	2		ns
Tcwh	CSN Inactive time	50		ns
Tcdz	CSN to Output High Z		42	ns

6 Register mapping table

Address (Hex)	Mnemonic	Bit	Reset Value	Type	Description
00	CONFIG				Configuration register
	Reserved	7	0	R/W	Retain, 0
	MASK_RX_DR	6	0	R/W	0: Receive interrupt enable, the RX_DR interrupt flag generates an interrupt signal on the IRQ pin, active low. 1: Receive interrupt disabled, the RX_DR interrupt flag does not affect the output of the IRQ pin.
	MASK_TX_DS	5	0	R/W	0: Transmit interrupt enabled, the TX_DS interrupt flag generates an interrupt signal on the IRQ pin, active low. 1: Transmit interrupt disabled, the TX_DS interrupt flag does not affect the output of the IRQ pin.
	MASK_MAX_RT	4	0	R/W	Maximum retransmission count interrupt mask control 0: Maximum retransmission count interrupt enabled, the MAX_RT interrupt flag generates an interrupt signal on the IRQ pin, active low 1: Maximum retransmission count interrupt disabled, the MAX_RT interrupt flag does not affect the output of the IRQ pin
	EN_CRC	3	1	R/W	Enable CRC. If EN_AA is not all zeros, EN_CRC must be 1. 0: Disable CRC 1: Enable CRC
	CRCO	2	0	R/W	CRC length configu ration, 0: 1 byte, 1: 2 bytes
	PWR_UP	1	0	R/W	Shutdown/Power-on Mode Configuration 0: Shutdown Mode 1: Power-on Mode
	PRIM_RX	0	0	R/W	The transmit/receive configuration can only be changed in Shutdown and Standby modes. 0: Transmit mode 1: Receive mode
01	EN_AA				Enable automatic confirmation
	Reserved	7:6	00	R/W	Retain, 0
	ENAA_P5	5	1	R/W	Enable Data Pipeline 5 to automatically confirm.

	ENAA_P4	4	1	R/W	Enable Data Pipeline 4 to automatically confirm.
	ENAA_P3	3	1	R/W	Enable automatic confirmation of Data Pipeline 3
	ENAA_P2	2	1	R/W	Enable automatic confirmation of Data Pipeline 2
	ENAA_P1	1	1	R/W	Enable automatic confirmation of Data Pipeline 1
	ENAA_P0	0	1	R/W	Enable automatic confirmation for Data Pipeline 0
02	EN_RXADDR				Enable the address of the data receiving pipeline
	Reserved	7:6	00	R/W	Retain, 0
	ERX_P5	5	0	R/W	Enable Data Pipeline 5
	ERX_P4	4	0	R/W	Enable Data Pipeline 4
	ERX_P3	3	0	R/W	Enable Data Pipeline 3
	ERX_P2	2	0	R/W	Enable Data Pipeline 2
	ERX_P1	1	1	R/W	Enable Data Pipeline 1
	ERX_P0	0	1	R/W	Enable data pipeline 0
03	SETUP_AW				Address width configuration
	Reserved	7:2	000000	R/W	Retain, 00000
	AW	1:0	11	R/W	Transmitter/Receiver Address Width 00: Error value 01:3 bytes 10:4 bytes 11:5 bytes
04	SETUP_RETR				Automatic retransmission configuration
	ARD	7:4	0000	R/W	Automatic retransmission delay configuration 0000:250 µs 0001: 500 µs 0010: 750 µs …… 1111: 4000 microseconds
	ARC	3:0	0011	R/W	Maximum number of automatic retransmissions 0000: Turn off auto-retransmission 0001: Once Train 0010:2 …… Train 1111:15
05	RF_CH				Radio frequency channel
	Reserved	7	0	R/W	Retain, 0
	RF_CH	6:0	0000010	R/W	Set the channel for the chip operation, corresponding to channels 0 to 125 respectively; the channel spacing is 1 MHz, and the default is 02, which is 2402 MHz.

06	RF_SETUP				Radio frequency configuration
	CONT_WAVE	7	0	R/W	When it is set to '1', the constant carrier wave transmission mode is enabled, which is used to test the transmission power.
	Reserved	6	0	R/W	Retain, 0
	RF_DR_LOW	5	0	R/W	Set the RF data rate to 250 kbps, 1 Mbps or 2 Mbps, and control it in conjunction with RF_DR_HIGH.
	PLL_LOCK	4	0	R/W	Reserved word, must be 0
	RF_DR_HIGH	3	1	R/W	Set the RF data rate [RF_DR_LOW, RF_ DR_HIGH]: 00: 1 Mbps 01: 2 Mbps 10: 250 kbps 11: Reserved
	RF_PWR	2:0	110	R/W	Set TX transmission power: 111: 7 dBm, 101: 3 dBm, 011: 0 dBm, 001: -6 dBm 110: 4dBm 100: 1dBm 010:-4dBm 000:-12dBm
07	STATUS				Status register (the first byte of SPI operation, the value of the status register is serially output through MISO).
	Reserved	7	0	R/W	Retain, 0
	RX_DR	6	0	R/W	RX FIFO has value flag bit, write '1' to clear.
	TX_DS	5	0	R/W	When the transmission of the interrupt bit at the transmitting end is completed, if it is in ACK mode, the TX_DS bit will be set to '1' upon receiving the ACK confirmation signal. Writing '1' to it will clear it.
	MAX_RT	4:3: 1	Zero one o ne	R/WR	When the maximum retransmission count is reached, the interrupt bit is set. Writing '1' to it clears the bit.
	RX_P_NO				The PPP number of the receiving pipeline that has received data can be read out through SPI. 000-101: Data pipelines 0-5 110: Not available 111: RX FIFO is empty
	TX_FULL	0	0	R	TX FIFO full flag bit.
08	OBSERVE_TX				Launch result statistics
	PLOS_CNT	7:4	0	R	Packet loss count. The maximum count is 15. After changing RF_CH, PLOS_CNT starts counting from 0.
	ARC_CNT	3:0	0	R	Re-transmission count. When a new packet is sent, ARC_CNT starts counting from 0.

09	RSSI				Received Signal Strength Detection
	Reserved	7:1	000000	R	
	RSSI	0	0	R	Received signal strength: 0: Received signal is less than -60 dBm
0A	RX_ADDR_P0	39:0	0xE7E7E7E7E7	R/W	The receiving address of data pipe 0 has a maximum width of 5 bytes (with the least significant byte written first, and the address width configured through SETUP_AW).
0B	RX_ADDR_P1	39:0	0xC2C2C2C2C2	R/W	The receiving address of data pipeline 1 has a maximum width of 5 bytes (with the least significant byte written first, and the address width configured through SETUP_AW).
0C	RX_ADDR_P2	7:0	0xC3	R/W	The lowest byte of the receiving address of data pipe 2 is the same as the highest byte of the receiving address of RX_ADDR_P1[39:8].
0D	RX_ADDR_P3	7:0	0xC4	R/W	The lowest byte of the receiving address of data pipe 3 is the same as the high byte of the receiving address, and is the same as RX_ADDR_P1[39:8].
0E	RX_ADDR_P4	7:0	0xC5	R/W	The lowest byte of the receiving address of data pipe 4 is the same as the high byte of the receiving address, and is identical to RX_ADDR_P1[39:8].
0F	RX_ADDR_P5	7:0	0xC6	R/W	The lowest byte of the receiving address of data pipe 5 is the same as the high byte of the receiving address, and it is also the same as RX_ADDR_P1[39:8].
10	TX_ADDR	39:0	0xE7E7E 7E7E7	R/W	The transmitting address of the transmitter (with the least significant byte written first), if the transmitter requires an ACK confirmation signal, the value of RX_ADDR_P0 needs to be configured to be equal to TX_ADDR and ARQ needs to be enabled. It should be noted that the most significant byte of the address cannot be set to 0xFF, 0x00, 0xA5, 0x5A, 0xAA, or 0x55, otherwise it may lead to receiving failure.
11	RX_PW_P0				
	Reserved	7:6	00	R/W	retain
	RX_PW_P0	5:0	0	R/W	The number of data bytes received by the data pipeline 0 (1 - 32 bytes). 1: 1 byte... 32: 32 bytes
12	RX_PW_P1				

	Reserved	7:6	00	R/W	retain
	RX_PW_P1	5:0	0	R/W	Number of bytes of data received by data pipeline 1 (1 - 32 bytes) 1:1 byte …… 32:32 bytes
13	RX_PW_P2				
	Reserved	7:6	00	R/W	retain
	RX_PW_P2	5:0	0	R/W	Receive data pipeline 2 data byte count (1 - 32 bytes). 1: 1 byte ······ 32: 32 bytes
14	RX_PW_P3				
	Reserved	7:6	00	R/W	retain
	RX_PW_P3	5:0	0	R/W	The number of data bytes received by the data pipeline 3 (1 - 32 bytes). 1: 1 byte ······ 32: 32 bytes
15	RX_PW_P4				
	Reserved	7:6	00	R/W	retain
	RX_PW_P4	5:0	0	R/W	Receive data pipeline 4 data byte count (1 - 32 bytes). 1: 1 byte ······ 32: 32 bytes
16	RX_PW_P5				
	Reserved	7:6	00	R/W	retain
	RX_PW_P5	5:0	0	R/W	Number of data bytes in the receiving data pipeline 5 (1 - 32 bytes) 1:1 byte …… 32:32 bytes
17	FIFO_STATUS				FIFO status
	Reserved	7	0	R/W	Retain, 0
	TX_REUSE	6	0	R	For the transmitter only, FIFO data is reused.

					When the REUSE_TX_PL command is used, the data that was successfully transmitted last time will be sent again. This function can be disabled by using the W_ TX_PAYLOAD or FLUSH TX command.
	TX_FULL	5	0	R	TX FIFO full flag: 1 - TX FIFO is full; 0 - TX FIFO is writable
	TX_EMPTY	4	1	R	TX FIFO empty flag: 1 - TX FIFO is empty; 0 - TX FIFO has data
	Reserved	3:2	00	R/W	Retain, 0
	RX_FULL	1	0	R	RX FIFO full flag: 1 - RX FIFO is full; 0 - RX FIFO is writable.
	RX_EMPTY	0	1	R	RX FIFO empty flag: 1 - RX FIFO is empty; 0 - RX FIFO has data
1C	DYNPD				Enable dynamic load length
	Reserved	7:6	0	R/W	Retain, 0
	DPL_P5	5	0	R/W	Enable dynamic load length for receive pipe 5 (requires EN_DPL and ENAA_P5).
	DPL_P4	4	0	R/W	Enable dynamic load length for receive pipe 4 (requires EN_DPL and ENAA_P4).
	DPL_P3	3	0	R/W	Enable dynamic load length for receive pipe 3 (requires EN_DPL and ENAA_P3).
	DPL_P2	2	0	R/W	Enable dynamic load length for receive pipe 2 (requires EN_DPL and ENAA_P2).
	DPL_P1	1	0	R/W	Enable dynamic load length for receive pipe 1 (requires EN_DPL and ENAA_P1).
	DPL_P0	0	0	R/W	Enable dynamic load length for receiving pipe 0 (requires EN_DPL and ENAA_P0).
1D	FEATURE			R/W	Feature register
	Reserved	7:3	0	R/W	Retain, 0000
	EN_DPL	2	0	R/W	Enable dynamic load length
	EN_ACK_PAY	1	0	R/W	Enable ACK load (ACK packets with load data)
	EN_DYN_ACK	0	0	R/W	Enable command W_TX_PAYLOAD_NOACK

7 Main parameter indicators

7.1 Extreme parameters

Working conditions	minimum value	maximum value	unit
Power supply voltage			
VDD	-0.3	3.6	V
VSS		0	V
Input voltage			
VI	-0.3	5.25	V
Output voltage			
VO	VSS to VDD	VSS to VDD	V
Total power consumption			
		100	mW
temperature			
Operating temperature range	-40	+85	℃
Storage temperature	-40	+125	℃
ESD performance	HBM(Human Body Model): Class 1C		

7.2 Electrical indicators

Conditions: VDD = 3V, VSS = 0V, TA = 27℃, Crystal Oscillator CL = 12pF

Symbols	parameters	minimum value	Typical value	maximum value	unit	Note
OP parameters						
VDD	Power supply voltage range	1.9		3.6	V	
I_{SHD}	Shutdown mode current		1		μA	
I_{STB}	Standby mode current		15		μA	
I_{IDLE}	Idle-TX mode current		380		μA	
I_{RX}@2MHZ	RX mode current @ 2 Mbps		15		mA	
I_{RX}@1MHZ	RX mode current @ 1 Mbps		14.5		mA	
I_{RX}@250kbps	RX@250kbps pattern current		14		mA	
I_{TX}@7dBm	TX@7dBm pattern current		25		mA	
I_{TX}@4dBm	TX@4dBm pattern current		16		mA	
I_{TX}@0dBm	TX@0dBm pattern current		12		mA	

I_{TX}@-6dBm	TX@-6dBm pattern current		9.5		mA	
I_{TX}@-12dBm	TX@-12dBm pattern current		8.5		mA	
RF parameters						
F_{OP}	RF frequency range	2400		2525	MHz	
F_{CH}	RF channel spacing	1			MHz	At 2 Mbps, it should be at least 2 MHz.
ΔF_{MOD} (2 Mbps)	Modulation frequency offset		±330		KHz	
ΔF_{MOD} (1M/250Kbps)	Modulation frequency offset		±175		KHz	
R_{GFSK}	Data rate	250		2000	Kbps	
RX parameters						
RX_{SENS}@2Mbps	Sensitivity @ 2 Mbps		-83		dBm	BER = 0.1%
RX_{SENS}@1Mbps	Sensitivity @ 1 Mbps		-87		dBm	BER = 0.1%
RX_{SENS}@250Kbps	Sensitivity @ 250 kbps		-96		dBm	BER = 0.1%
C/I_{CO}@2Mbps	Co-channel selectivity		6		dB	
C/I_{1st}@2Mbps	First adjacent channel selectivity 2 MHz		0		dB	
C/I_{2ND}@2Mbps	Second adjacent channel selectivity 4 MHz		-20		dB	
C/I_{3RD}@2Mbps	Third adjacent channel selectivity 6 MHz		-26		dB	
C/I_{CO}@1Mbps	Co-channel selectivity		7		dB	
C/I_{1st}@1Mbps	First adjacent channel selectivity 2 MHz		6		dB	
C/I_{2ND}@1Mbps	Second adjacent channel selectivity 4 MHz		-21		dB	
C/I_{3RD}@1Mbps	Third adjacent channel selectivity 6 MHz		-30		dB	
TX parameter						
P_{RF}	RF output power	-30		7	dBm	
P_{BW}@2Mbps	Modulation bandwidth		2.1		MHz	
P_{BW}@1Mbps	Modulation bandwidth		1.1		MHz	
P_{BW}@250Kbps	Modulation bandwidth		0.9		MHz	
P_{RF1}	First adjacent channel power 2 MHz			-20	dBm	
P_{RF2}	Second adjacent channel power 4 MHz			-46	dBm	
Crystal oscillator parameters						
F_{XO}	Crystal oscillator frequency		16		MHz	
Delta F	Frequency offset		±60		ppm	
ESR	Equivalent loss resistance		100		Omega	

8 Encapsulation

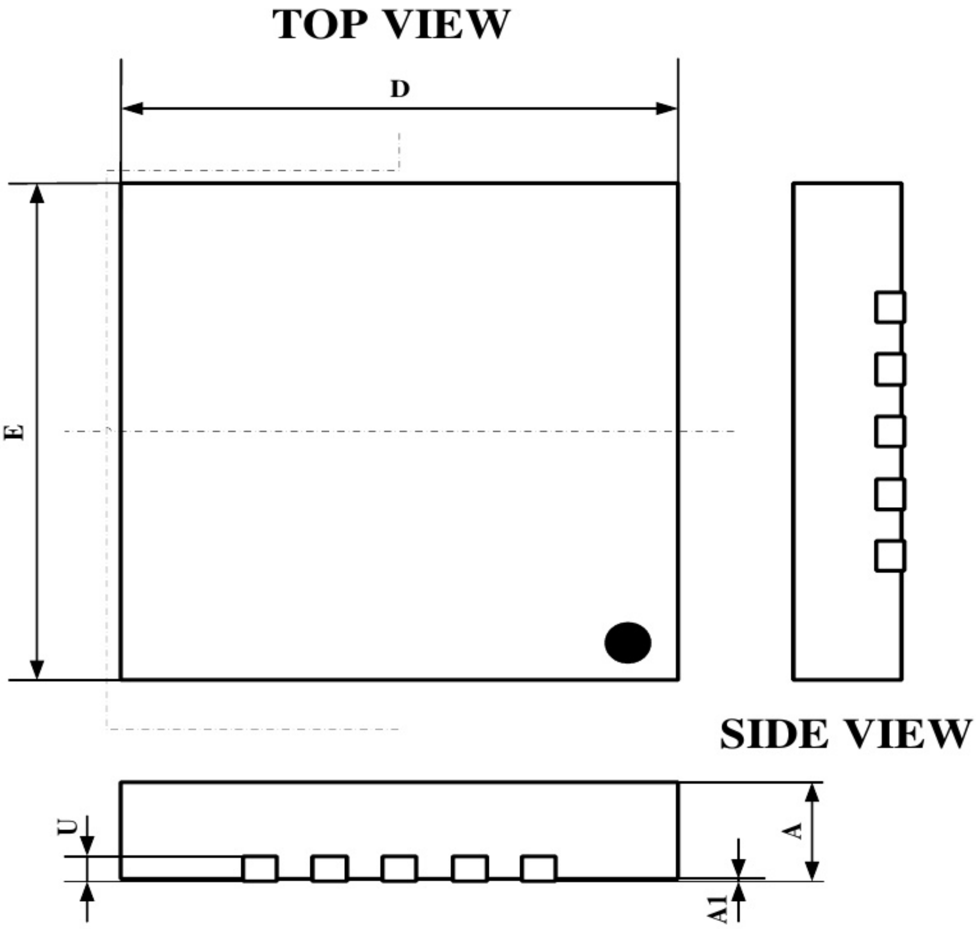

Figure 8-1 Top-level Diagram

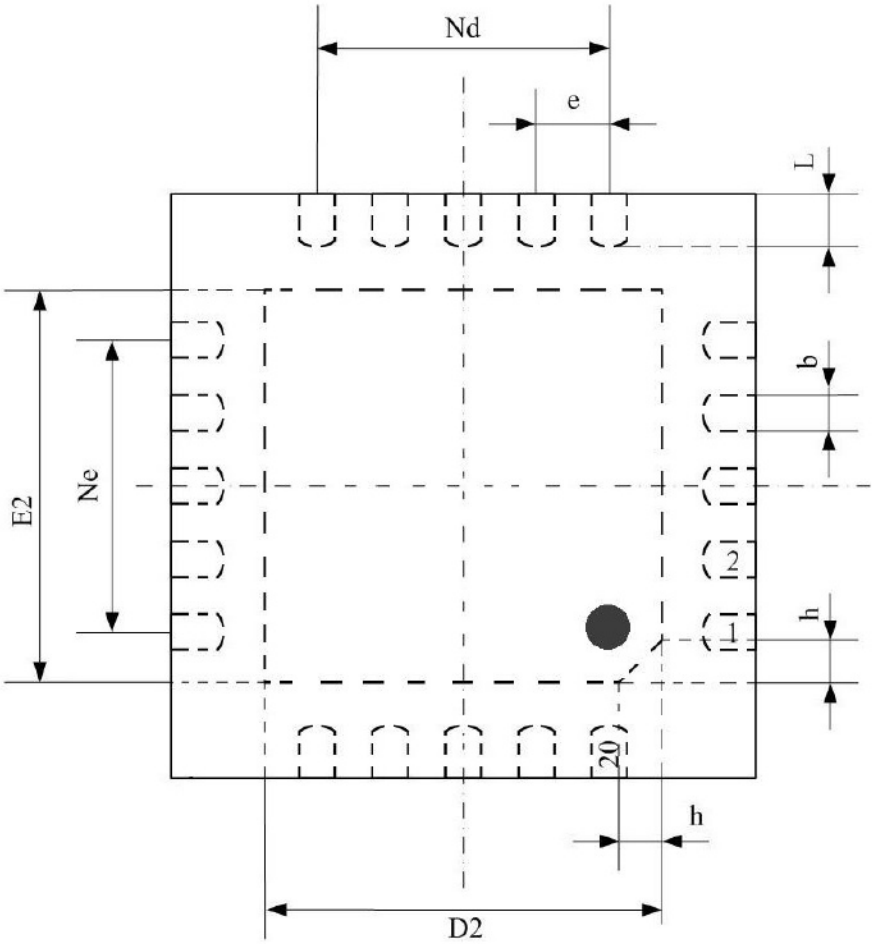

Figure 8-2 Encapsulation Dimensions (Top View)

SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.70	0.75	0.80
A1	—	0.02	0.05
b	0.18	0.25	0.30
D	3.90	4.00	4.10
D2	2.55	2.65	2.75
e	0.50 BSC		
E2	2.55	2.65	2.75
E	3.90	4.00	4.10
Ne	2.00 BSC		
Nd	2.00 BSC		
L	0.35	0.40	0.45
h	0.30	0.35	0.40
U	0.20 REF.		
L/F Carrier Size (mil)	114 × 114		

9 Typical application schematic diagram

9.1 Typical application schematic diagram

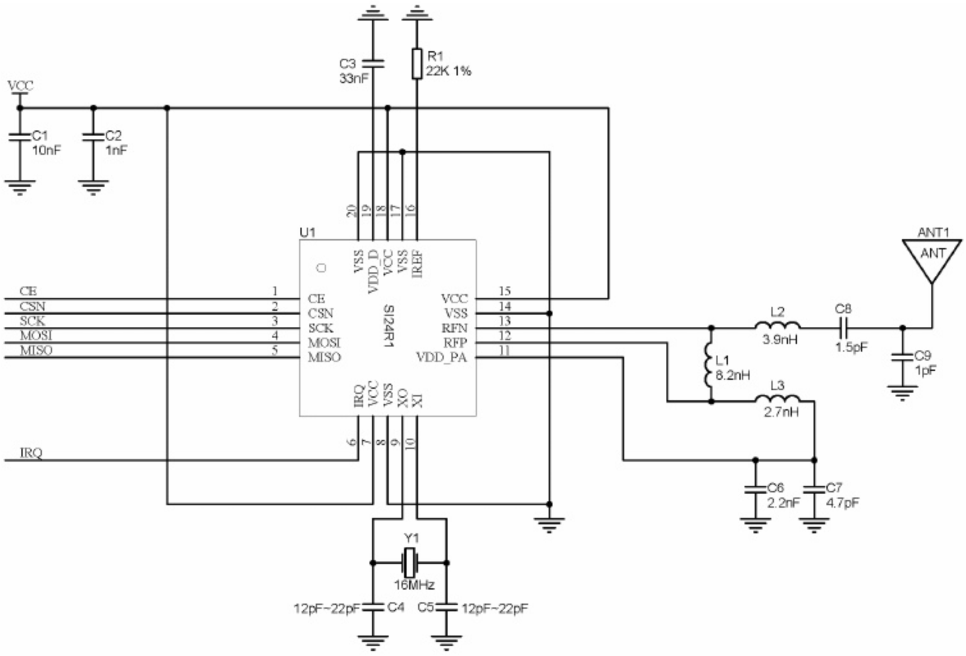

Figure 9-1 Typical Application Schematic Diagram

Table 9-1 Bill of Materials for Components

Device Name	value	form; shape; format	Description
C1	100uF	0402	X7R, +/- 10%
C2	1nF	0402	X7R, +/- 10%
C3	33nF	0402	X7R, +/- 10%
C4	12~22pF	0402	NPO, +/- 2%
C5	12~22pF	0402	NPO, +/- 2%
C6	2.2 nanofarads	0402	X7R, +/- 10%
C7	4.7 pF	0402	NPO, ± 0.25 pF
C8	1.5 pF	0402	NPO, ± 0.1 pF
C9	1.0 pF	0402	NPO, ± 0.1 pF
L1	8.2 nH	0402	chip inductor, +/- 5%
L2	3.9 nanohenries	0402	chip inductor, +/- 5%
L3	2.7 nanohenries	0402	chip inductor, +/- 5%
R1	22KΩ	0402	plus or minus 1%
R2	Not mouted	0402	
Y1	16MHz		+/-60ppm, CL=12pF
U1		QFN20 04×04	

When the system cannot provide a stable power supply voltage, for example, when powered by a button battery, it is recommended that the system use a 100uF capacitor to stabilize the power supply voltage. At the same time, it should be noted that the capacitor should not have a large leakage current.

The pins CE, CSN, SCK, MOSI, MISO, and IRQ are the interfaces with the MCU. When the MCU is not operating the Si24R1, the output pins MISO and IRQ are left floating, and the input pins CE, CSN, SCK, and MOSI must be connected to the power supply or ground through the MCU interface.

9.2 PCB Routing

The PCB layout shown in the following figure is an example of the PCB layout for the typical circuit diagram mentioned above. Here, all PCB boards are FR-4 double-sided boards, with a copper layer on both the top and bottom sides. The copper layers on the top and bottom are connected by a large number of vias, while there is no copper layer beneath the antenna. The bottom of the chip is grounded. To ensure better RF performance, it is recommended that the exposed die at the bottom of the chip be connected to a large area of the ground on the PCB.

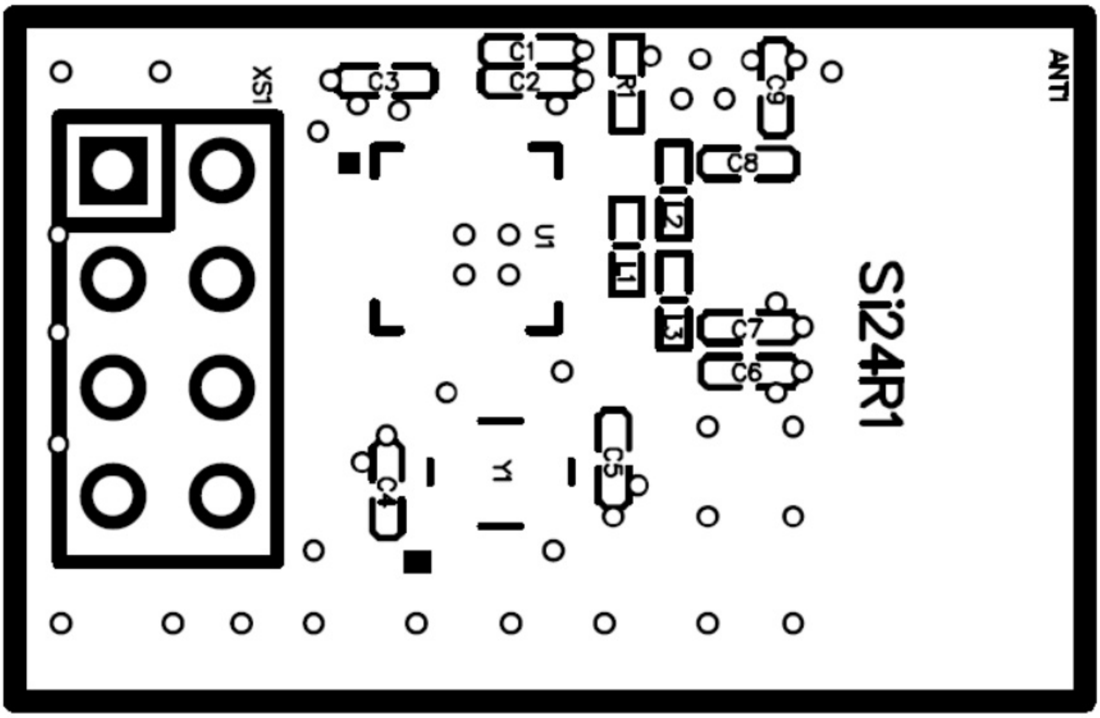

Figure 9-2 Top-layer silk screen diagram of on-chip antenna (0402 components)

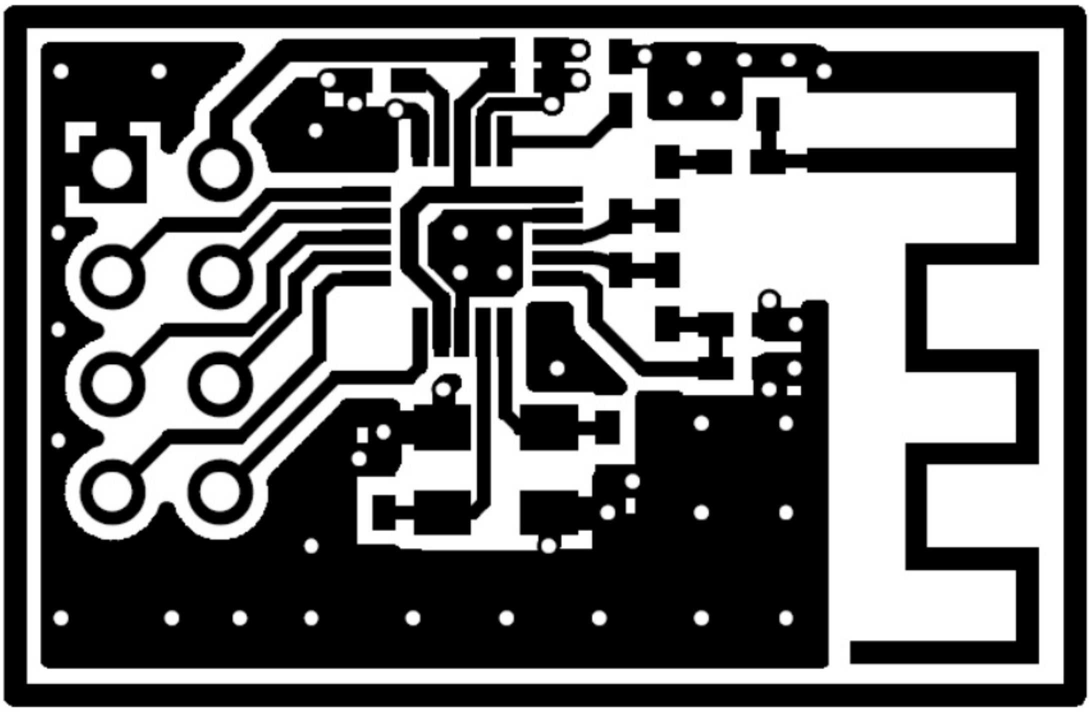

Figure 9-3 Top-layer Wiring Diagram of On-Chip Antenna (0402 Components)

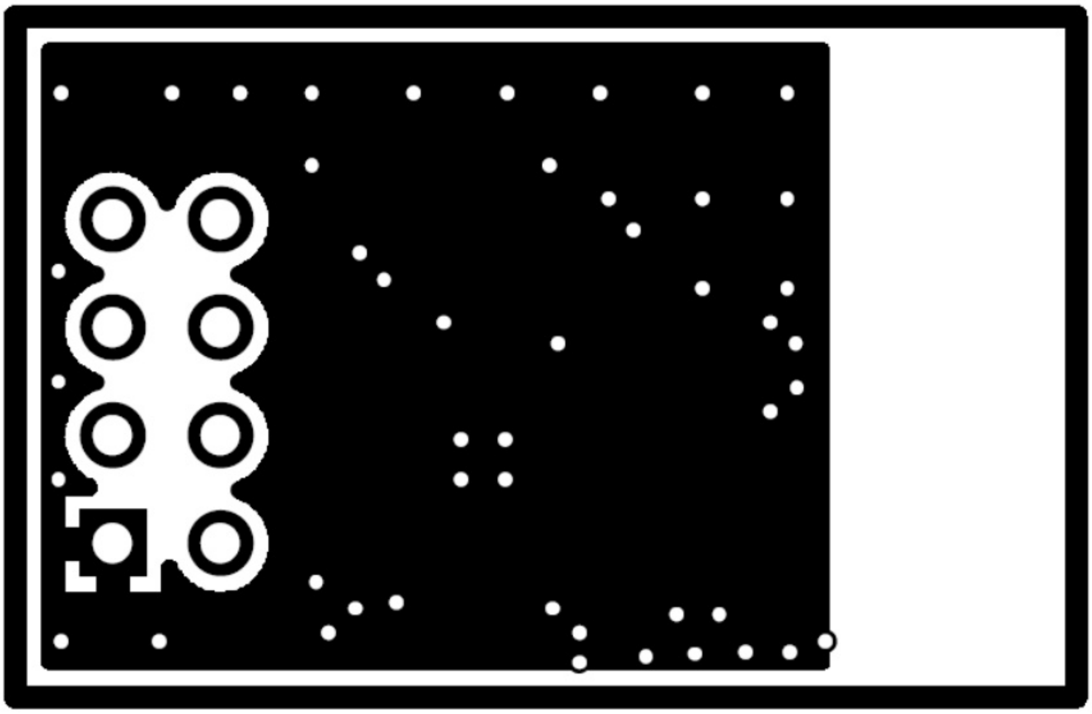

Figure 9-4 Underlying Wiring Diagram of On-Chip Antenna

10 Version Information

Version	Modification date	Amendments to the content
Version 1.0	2021/12/02	Modify contact information
Version 1.1	2022/10/20	Chapter 9: Modifying the BOM Table of the Schematic Diagram and Adding Package Introduction
Version 1.2	2022/10/24	Modify order information
Version 1.3	2024/03/19	Add the description of the ARQ packet format and the address related to register 0x10.
Version 1.4	2024/04/03	Description of the PID for modifying the data packet format
Version 1.5	2024/06/14	Modify the pin diagram

11 Order Information

Encapsulation mark

Si24R1 ABBCDEE

Si24R1+: Chip code

A: The year code for the packaging date, 5 represents the year 2020.

BB: Processed shipment weekly report, for example, 42 represents the 42nd week of year A for processing. C: Packaging factory code, which can be A, HT, NJ or WA, and is also abbreviated as A, H, N or W. D: Testing factory code, which can be A, Z or H.

EE: Production batch code

Table 11-1 Order Information Table

Order code	Encapsulation	Packaging	smallest unit
Si24R1-Sample	4×4mm 20-pin QFN	Box/Tube	5
Si24R1	4×4mm 20-pin QFN	Tape and reel	4K

12 Technical Support and Contact Information

Technical Support Center of Nanjing Institute of Microelectronics Technology, Chinese Academy of Sciences

Telephone: 025-68517780

Address: Room 201, Building B, R&D Zone 3, Xuzhuang Software Park, Xuanwu District, Nanjing City

Website: http://www.csm-ic.com

Marketing and sales

Mobile: 13645157034, 13645157035Email: sales@
csmic.ac.cn

Technical support

Mobile phone: 13645157034

Email: supports@csmic.ac.cn

Typical Configuration Scheme

Mode One: **ACK** Communication

Transmitter configuration:

spi_rw_reg(SETUP_AW, 0x03); // Set the address width to 5 bytes

spi_write_buf(TX_ADDR, TX_ADDRESS, 5); Write the sending address, 5 bytes.

spi_write_buf(RX_ADDR_P0, TX_ADDRESS, 5); The receiving channel 0 address and the transmitting address are the same. spi_write_buf(W_TX_PAYLOAD, buf, TX_PLOAD_WIDTH); // Write to TX FIFO

spi_rw_reg(FEATURE, 0x04); Enable dynamic load length

spi_rw_reg(DYNPD, 0x01); // Enable DPL_P0

spi_rw_reg(SETUP_RETR, 0x15); The automatic retransmission delay wait time is 500 microseconds, and the automatic retransmission is performed 5 times.

spi_rw_reg(RF_CH, 0x40); Select the RF channel

spi_rw_reg(RF_SETUP, 0x0e); transmission rate: 2 Mbps and power

spi_rw_reg(CONFIG, 0x0e); Configure for transmission mode, CRC, and maskable interrupts.

CE = 1; Data

Receiver configuration:

spi_write_buf(RX_ADDR_P0, TX_ADDRESS, 5); spi_rw_reg(EN_RXADDR, 0x01); // The address of receiving channel 0 is the same as the transmitting address. Enable receiving channel 0.

spi_rw_reg(RF_CH, 0x40); Select the RF channel

spi_rw_reg(SETUP_AW, 0x03); // Set the address width to 5 bytes

spi_rw_reg(FEATURE, 0x04); Enable dynamic load

spi_rw_reg(DYNPD, 0x01); // Enable DPL_P0

spi_rw_reg(RF_SETUP, 0x0e); The data transmission rate is 2 Mbps and the power spi_rw_reg (CONFIG, 0x0f); // Set to receive mode, CRC, and maskable interrupt.

CE = 1;

Mode Two: NOACK Communication

Transmitter configuration:

spi_write_buf(TX_ADDR, TX_ADDRESS, 5); spi_rw_reg(FEATURE, 0x01); // Write the sending address and enable the W_TX_PAYLOAD_NOACK commandspi_write_buf(W_TX_PAYLOAD_NOACK, buf, TX_PLOAD_WIDTH); // Write to the FIFOspi_rw_reg(SETUP_AW, 0x03); // 5 byte Address width

spi_rw_reg(RF_CH, 0x40); Select RF channel 0x40

spi_rw_reg(RF_SETUP, 0x08); Data transfer rate: 2 Mbps

spi_rw_reg(CONFIG, 0x0e); Configure as transmission mode, CRC is 2 bytes.

CE = 1;

Receiver configuration:

spi_write_buf(RX_ADDR_P0, TX_ADDRESS, 5); // Receive address spi_rw_reg(EN_
RXADDR, 0x01); // Enable receive channel 0

spi_rw_reg(RF_CH, 0x40); Select the RF channel

spi_rw_reg(RX_PW_P0, TX_PLOAD_WIDTH); // Set the data width of receive channel 0 to load spi_rw_reg(RF_
SETUP, 0x08); // Data transmission rate 2Mbps, -18dbm TX power spi_rw_reg(CONFIG, 0x0f); // Configure as receiver, CRC 2 bytes

CE = 1;

Mode Three: The recipient opens multiple channels.

Dynamic load:

```
spi_rw_reg(FEATURE, 0x04);
spi_rw_reg(DYNPD, 0x3F) ;          Enable dynamic load length for all channels
spi_rw_reg(EN_RXADDR, 0x3F);           Open all channels.
spi_rw_reg(RF_CH, 0x40);           RF channel 0x40
spi_rw_reg(SETUP_AW, 0x03);         // 5 byte Address width
spi_rw_reg(CONFIG, 0x0B);          Configure as the receiving party.
CE = 1;
```

Static load:

```
spi_rw_reg(RX_PW_P0, 0x20);         // Set the data width for channel 0 to receive data
spi_rw_reg(RX_PW_P1, 0x20);
spi_rw_reg(RX_PW_P2, 0x20);
spi_rw_reg(RX_PW_P3, 0x20);
spi_rw_reg(RX_PW_P4, 0x20);
spi_rw_reg(RX_PW_P5, 0x20); Select

spi_rw_reg(EN_RXADDR, 0x3F);           Open all channels.
spi_rw_reg(RF_CH, 0x40);            Select RF channel 0x40
spi_rw_reg(SETUP_AW, 0x03);         // Set address width
spi_rw_reg(CONFIG, 0x0F);          Configure as the receiving party.
CE = 1;
```